

Course Title	System and Network Administration Lab
Course Code	DI-323L
Credit Hours	1
Category	Domain Core
Prerequisite	CC-311: Operating Systems
Co-Requisite	None
Follow-up	None
Course Description	Introduction: Review of computer networks lab, Server Environment (Microsoft and Linux). Installation and Configuration: Client Server OS Configuration, Installation and configuration of servers, Linux Installation and Verification, Installation of Activity Directory on Domain Controller, Manage and Active Directory Forest and domain. Managing Network Services: Network Monitoring Tools. Boot Management and Process Management. IP Tables and Filtering. Securing Network Traffic. Advanced File Systems and Logs. Bash Shell Scripting. Managing Servers: FTP, NFS, Samba, DHCP, DNS and Apache. Configuring Local Services and Managing Basic System Issues. Administer Users and Groups. Domain controller. Installation and Configuration of Wireless Access Points: Installation and configuration of wireless Router, Manageable and non-manageable Cisco switches, Concept and configuration of VLAN, Switch modes and operations, setting up WAN on simulator, identifying necessary devices to build a WAN, learning static and dynamic routing protocols, Understanding and implementing RIP (Routing Information Protocol). Understanding and implementing IGRP, ACL's configuration on routers.
Text Book(s)	1. The Practice of System and Network Administration, Second Edition by Thomas Limoncelli, Christina Hogan and Strata Chalup, Addison-Wesley Professional; 2nd Edition (2007). ISBN-10: 0321492668. 2. Study guide for Practice of System and Network Administration by Thomas A. Limoncelli, Cram101; 2nd Edition (2011). ISBN-10: 1428851755. 3. Networking Systems Design and Development by Lee Chao, CRC Press; 1st Edition (December 21, 2009). ISBN-10: 142009159X (TB2).
Reference Material	1. Red Hat Enterprise Linux 6 Bible: Administering Enterprise Linux Systems by William von Hagen, 2011.

Curricula/Syllabi of BS Information Technology for Punjab University Affiliated Colleges

Course Title	System and Network Administration
Course Code	DI-323
Credit Hours	3+1
Category	Domain Core
Prerequisite	CC-311: Operating Systems
Co-Requisite	None
Follow-up	None
Course Description	Introduction: System Administration (SA), SA Components, Server Environment (Microsoft and Linux), Reliable Products, Server Hardware Costing, Maintenance Contracts and Spare Parts, Maintaining Data Integrity, Client Server OS Configuration, Providing Remote Console Access. Comparative Analysis of OS: Important Attributes, Key Features, Pros and Cons. Linux Installation and Verification: Configuring Local Services and Managing Basic System Issues. Administer Users and Groups. System and Network Management: Software Management. Managing Network Services and Network Monitoring Tools. Boot Management and Process Management. IP Tables and Filtering. Securing Network Traffic. Advanced File Systems and Logs. Bash Shell Scripting. Configuring Servers (FTP, NFS, Samba, DHCP, DNS and Apache).
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